



DCA0305

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Advanced RAG

How the traditional RAG pipeline was improved?

Where we are in the LLM block

05

History of LLMs

From word counting to agents

06

Inside the Transformer

GPT-2 in seven stops

07

Ollama and local models

Team radio classifier

08

Classic RAG

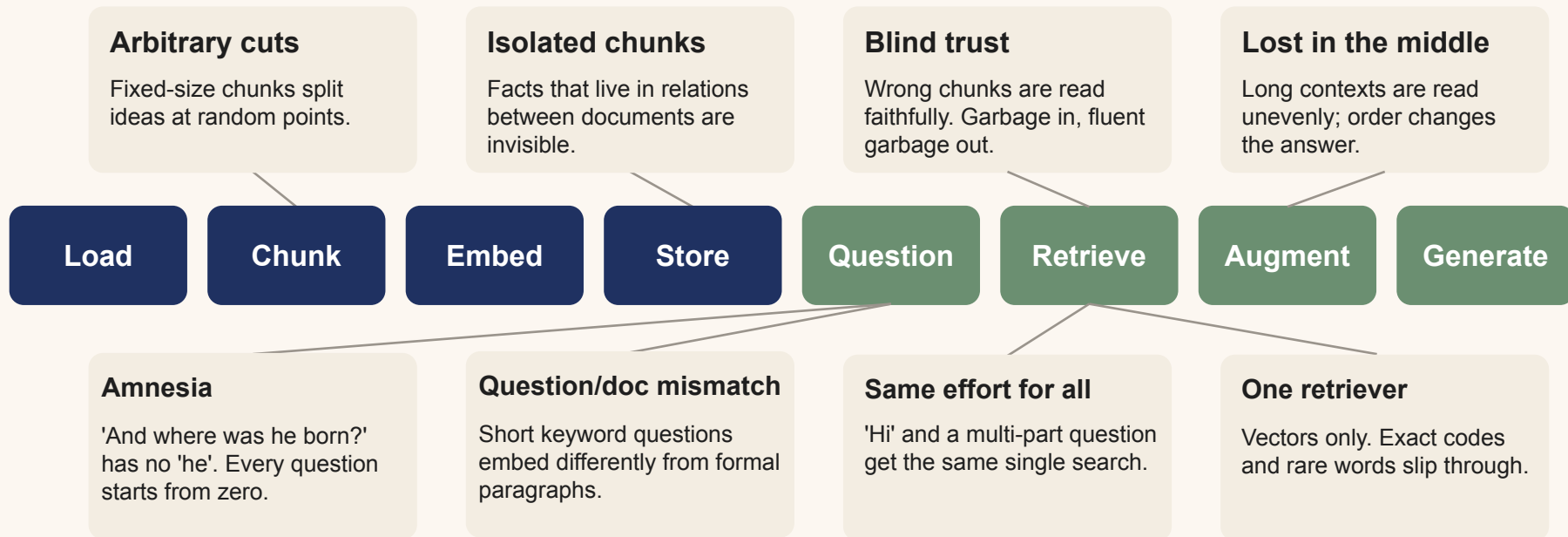
Your documents in the prompt

09

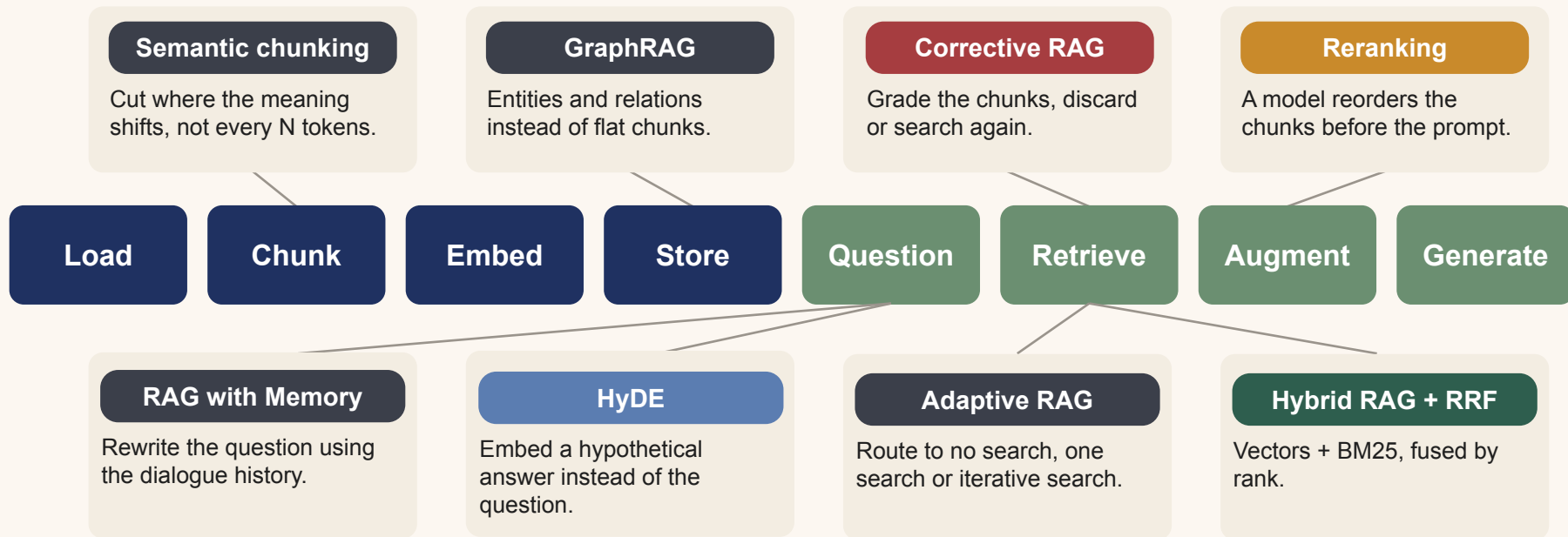
Advanced RAG

Today · HyDE, hybrid, reranking, CRAG

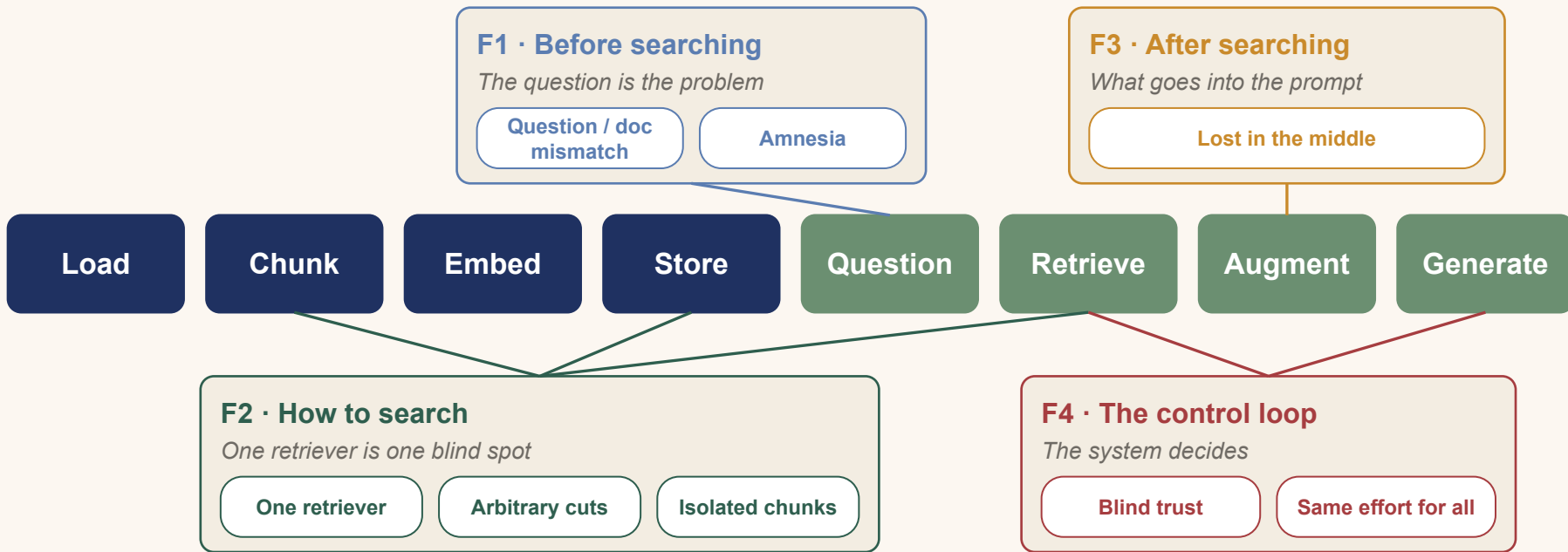
Where traditional RAG breaks

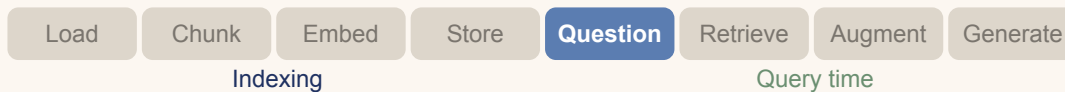


The RAG family



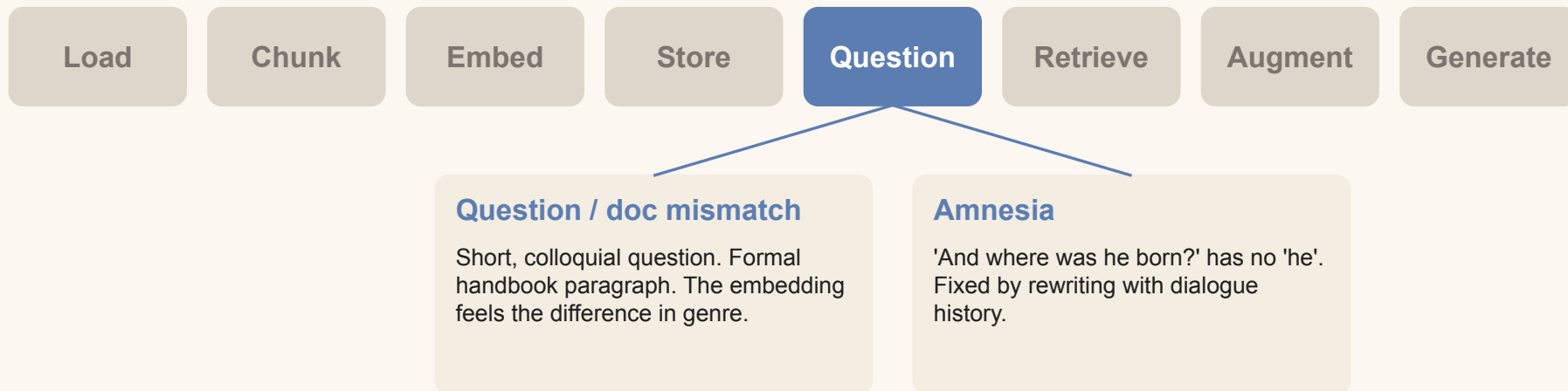
Grouping in four families





F1 · Before searching

The question is the problem



Load

Chunk

Embed

Store

Question

Retrieve

Augment

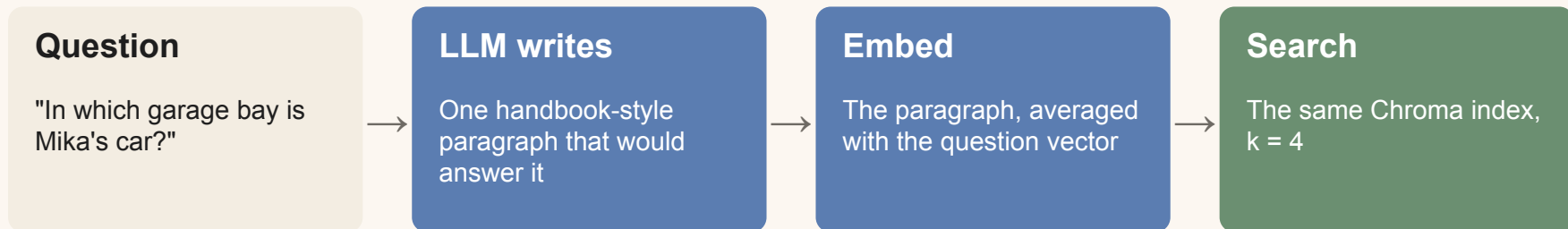
Generate

Indexing

Query time

Hypothetical Document Embeddings (HyDE)

Gao et al., 2023 · Precise Zero-Shot Dense Retrieval without Relevance Labels



It may be wrong on the numbers and still help.

The hypothetical paragraph is never shown to the user and never enters the prompt. It only speaks the handbook's language, so it lands near the right chunk. **The price is one extra generation per question.**

Load

Chunk

Embed

Store

Question

Retrieve

Augment

Generate

Indexing

Query time

What the notebook shows

RESULT 1

Original query and Hypothetical query

Query: In which garage bay is Mika Sørensen's car, seen from the pit lane?

Hypothetical query: In accordance with the established procedures of the Aurora Racing team, the position of Mika Sørensen's car within the garage bay is meticulously documented and adhered to the Vehicle Layout Procedure. For the specific instance of Mika Sørensen's car, the vehicle is always parked in Garage Bay 3, as per the Vehicle Layout Procedure, ensuring that all team members, including the pit crew, can access the car efficiently and effectively. This standardized approach not only facilitates quick and accurate vehicle preparation but also underscores the Aurora Racing team's commitment to precision and efficiency in all aspects of our operations.

qwen2.5:3b writes

nomic-embed-text

k = 4

Load

Chunk

Embed

Store

Question

Retrieve

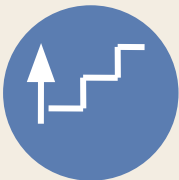
Augment

Generate

Indexing

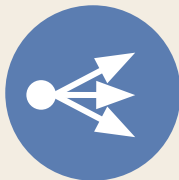
Query time

Same family: rewrite the question first



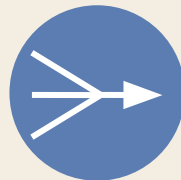
Step-Back

Ask the principle before the detail. 'Which rule governs tyre pressure in the wet?' then the specific question.



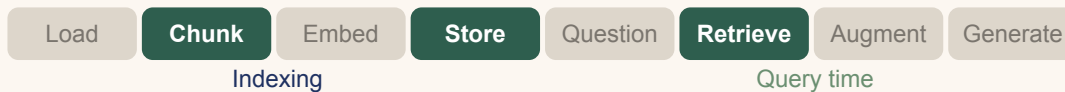
Multi-query

Generate N reformulations, search each one, merge the lists.



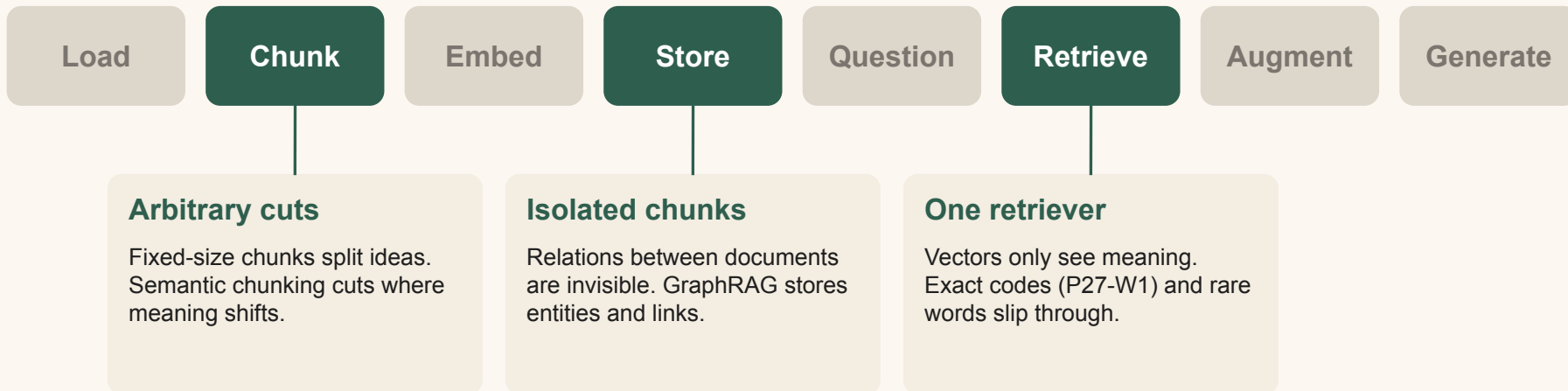
RAG-Fusion

Multi-query with the lists fused by Reciprocal Rank Fusion (RRF).



F2 · How to search

One retriever is one blind spot



Load

Chunk

Embed

Store

Question

Retrieve

Augment

Generate

Indexing

Query time

Semantic chunking

Cut where the meaning shifts, not every N tokens · the fix for arbitrary cuts

1 · Split

The page into sentences

2 · Embed

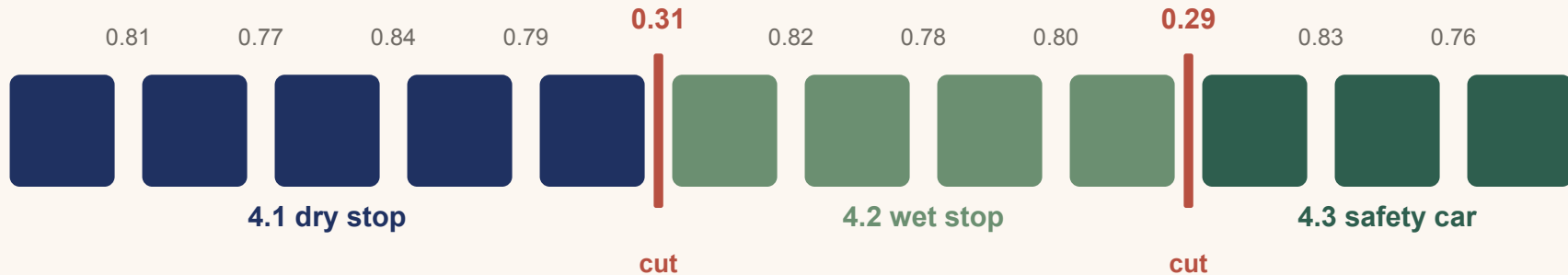
One vector per sentence,
the same model as the
index

3 · Compare

Cosine between each
sentence and the next

4 · Cut

Where the cosine drops
below a threshold



Load

Chunk

Embed

Store

Question

Retrieve

Augment

Generate

Indexing

Query time

BM25: counting tokens, weighted

$$\text{score}(q, d) = \sum_{t \in q} \text{idf}(t) \cdot \frac{\text{tf}(t, d) (k_1 + 1)}{\text{tf}(t, d) + k_1 \left(1 - b + b \frac{|d|}{\text{avgdl}} \right)}$$

$\text{idf}(t) = \ln \left(\frac{N - n_t + 0.5}{n_t + 0.5} + 1 \right)$ a rare term weighs more. N chunks in the corpus, n_t of them contain t

$\text{tf}(t, d)$ occurrences of t in the chunk, saturated by k_1 : the tenth repeat adds almost nothing

$|d|, \text{avgdl}$ chunk length against the average; b sets how much a long chunk is discounted

$k_1 \approx 1.5, b = 0.75$ the usual defaults (rank_bm25 ships $k_1 = 1.5, b = 0.75$)

question: "... in procedure
P27-W1?"

tokens: procedure p27-w1 tyres used

p27-w1: 2 of 14 chunks → high idf
procedure: 9 of 14 → low idf

The register of codes wins on the rare token. The dense retriever never saw a token; it saw a direction.

Load

Chunk

Embed

Store

Question

Retrieve

Augment

Generate

Indexing

Query time

Hybrid: fuse by rank, not by score

Reciprocal Rank Fusion · as used in RAG-Fusion, Rackauckas, 2024

$$\text{RRF}(d) = \sum_{r \in R} \frac{1}{k + \text{rank}_r(d)} \quad k = 60$$

 R

the set of ranked lists being fused (here two: dense and BM25)

 $\text{rank}_r(d)$

position of chunk d in list r , starting at 1; absent means no contribution

 k

smoothing constant; 60 keeps a first place from dominating the sum

Dense list (cosine)

1. A 0.82 → 0.0164

2. B 0.79 → 0.0161

3. C 0.71 → 0.0159

BM25 list (score)

1. C 7.3 → 0.0164

2. A 4.1 → 0.0161

3. D 2.6 → 0.0159

Fused (sum of both lists)

1. A 0.0164 + 0.0161 = 0.0325

2. C 0.0159 + 0.0164 = 0.0323

3. B 0.0161

4. D 0.0159

Load

Chunk

Embed

Store

Question

Retrieve

Augment

Generate

Indexing

Query time

What the notebook shows

RESULT 2

Dense, BM25 and the fused list, side by side

Q: Which blanket temperature applies to the tyres used in procedure P27-W1?

#	dense (cosine similarity)	BM25 (score)	hybrid (RRF)
1	handbook_tyres · 2.2 Reference pressures (0.740)	procedure_codes · §9 Procedure register (8.78)	procedure_codes · §9 Procedure register (0.03202)
2	handbook_procedures · 4.1 Dry-race pit stop (0.686)	handbook_procedures · 4.2 Wet-race pit stop (6.29)	handbook_procedures · 4.1 Dry-race pit stop (0.03200)
3	handbook_procedures · 4.2 Wet-race pit stop (0.680)	handbook_procedures · 4.1 Dry-race pit stop (5.60)	handbook_procedures · 4.2 Wet-race pit stop (0.03200)
4	procedure_codes · §9 Procedure register (0.666)	strategy_playbook · 6.2 Overcut (4.38)	handbook_tyres · 2.2 Reference pressures (0.03154)

Load

Chunk

Embed

Store

Question

Retrieve

Augment

Generate

Indexing

Query time

Same family: a better index



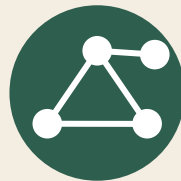
Semantic chunking

Shown earlier today. Cut where the similarity between neighbouring sentences drops.



Parent-child

Search small chunks for precision, hand the parent section to the prompt for context.



GraphRAG

Extract entities and relations, retrieve subgraphs.

Load

Chunk

Embed

Store

Question

Retrieve

Augment

Generate

Indexing

Query time

F3 · After searching

What goes into the prompt

Load

Chunk

Embed

Store

Question

Retrieve

Augment

Generate

Lost in the middle

k = 12 fills the prompt with near-duplicates and off-topic text. A 3B model reads the middle worst of all.

Load

Chunk

Embed

Store

Question

Retrieve

Augment

Generate

Indexing

Query time

Bi-encoder vs cross-encoder

Bi-encoder (embedding)

Question

↓ vector

Chunk

↓ vector

Two texts, two towers, one cosine. Vectors are computed once and stored. Fast enough to index the whole corpus.

Cross-encoder (reranker)

Question + chunk → one score

Reads the pair together, attends across both. Nothing to store, one forward pass per pair. Too slow for the corpus, ideal for a dozen candidates.

Retrieve 12



Rerank



keep 4, best first

Load

Chunk

Embed

Store

Question

Retrieve

Augment

Generate

Indexing

Query time

What the notebook shows

RESULT 3

The bi-encoder order and the cross-encoder order

Q: Why does car 88 use a higher rear tyre pressure than car 27 in the wet?

final	bi-encoder	cross-encoder	chunk
1	7	6.54	drivers.md · Car 88 · Mika Sørensen
2	1	6.11	handbook_procedures.md · 4.2 Wet-race pit stop
3	4	1.22	handbook_procedures.md · 4.1 Dry-race pit stop
4	5	-0.98	garage_and_pitlane.md · §10 Garage and pit lane

Load

Chunk

Embed

Store

Question

Retrieve

Augment

Generate

Indexing

Query time

Same family: shape the context



Context compression

Keep only the sentences of each chunk that answer the question. Fewer tokens, same evidence.



Ordering

Best chunk first and last, weakest in the middle. The model reads the edges; the middle is where facts get lost.

Load

Chunk

Embed

Store

Question

Retrieve

Augment

Generate

Indexing

Query time

F4 · The control loop

the system decides

Load

Chunk

Embed

Store

Question

Retrieve

Augment

Generate

Blind trust

The generator reads whatever arrives with the same faith. The grounding rule was the only defence.

Same effort for all

'Hi' and a three-part question get one search each. Adaptive RAG routes by question type.

Load

Chunk

Embed

Store

Question

Retrieve

Augment

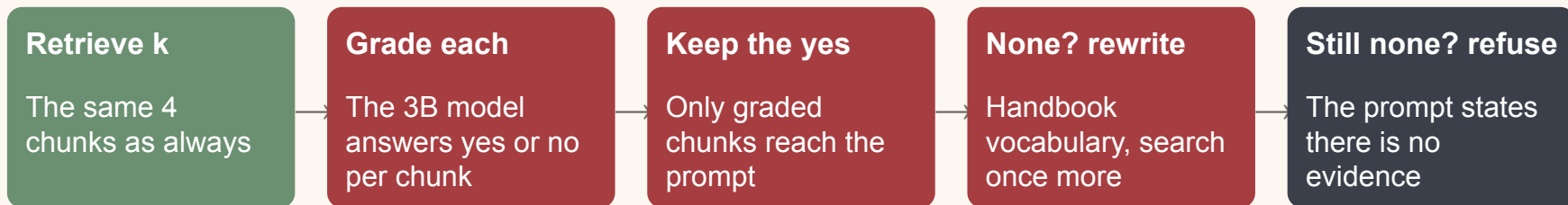
Generate

Indexing

Query time

Corrective RAG

Yan et al., 2024 · A grader between the search and the prompt



The grader is the same generator, with the one-word contract from lesson 07.

Price: up to 4 gradings, plus 1 rewrite and 4 more gradings, plus the answer. Up to 10 LLM calls for one question. **What it**

buys: an unanswerable question no longer reaches the generator with four irrelevant chunks and a prayer.

Load

Chunk

Embed

Store

Question

Retrieve

Augment

Generate

Indexing

Query time

What the notebook shows

RESULT 4

The grader on two questions, then the loop on all 20

d01 · What tyre pressure do we use for car 27 in the wet?

verdict	chunk
✓ yes	handbook_procedures.md · 4.2 Wet-race pit stop
✓ yes	handbook_tyres.md · 2.2 Reference pressures
✗ no	handbook_procedures.md · 4.4 Red flag
✓ yes	handbook_procedures.md · 4.1 Dry-race pit stop

Load

Chunk

Embed

Store

Question

Retrieve

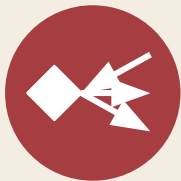
Augment

Generate

Indexing

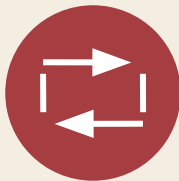
Query time

Same family: the system decides



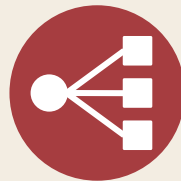
Adaptive RAG

A router picks no search, one search or iterative search by question type.



Self-RAG

The model critiques its own draft (supported? useful?) and revises.



Agentic RAG

An LLM decides which of today's tools to call and when.

Load

Chunk

Embed

Store

Question

Retrieve

Augment

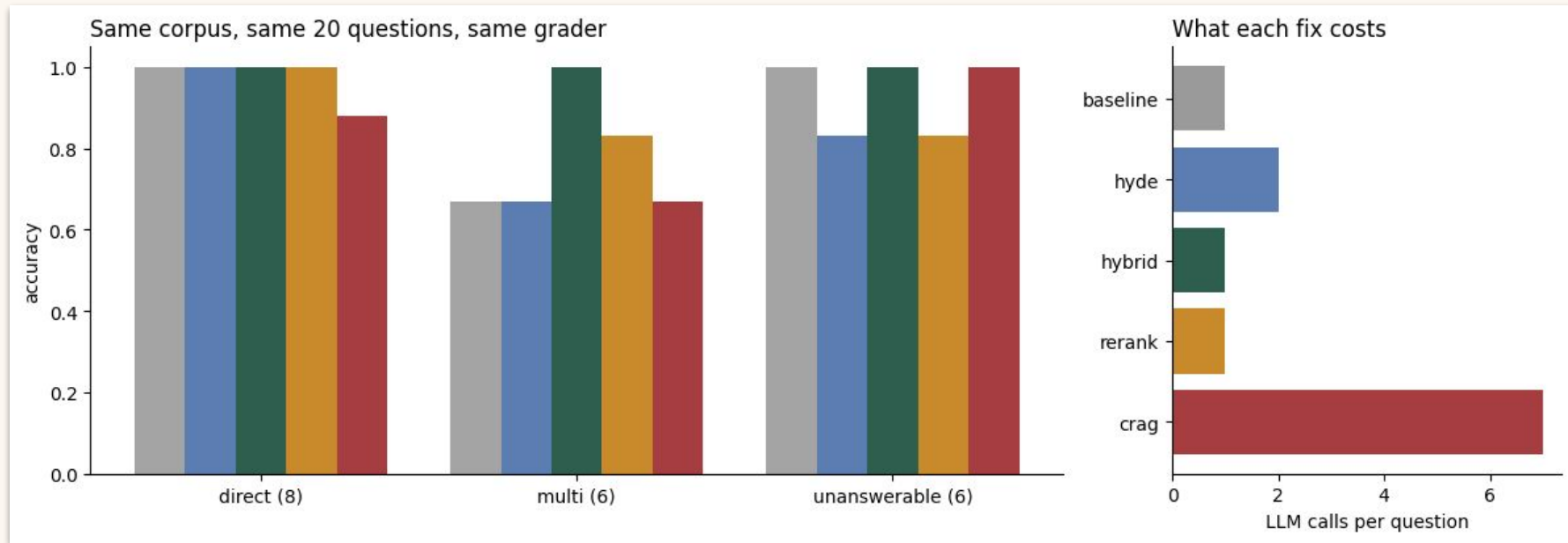
Generate

Indexing

Query time

Scoreboard: four patches against the baseline

Same corpus, same 20 questions, same grader



Baseline

Hyde

Hybrid

Rerank

Crag

Load

Chunk

Embed

Store

Question

Retrieve

Augment

Generate

Indexing

Query time

Did each fix earn its complexity?

	LLM calls / question	Extra model	Latency (s, mean)	Accuracy (20 q) and gain
baseline	1.0	None	0.93	0.85
HyDE	2.0 (+1, the paragraph)	None	2.53	0.85 (0.00)
Hybrid + RRF	1.0	BM25 index	1.01	1.00 (+0.15)
Reranking	1.0	Cross-encoder, 22M	1.14	0.90 (+0.05)
CRAG	7.0	None	1.02	0.85 (0.00)

[Load](#)[Chunk](#)[Embed](#)[Store](#)[Question](#)[Retrieve](#)[Augment](#)[Generate](#)

Indexing

Query time

This week

Notebook · lesson09_exercise.ipynb

Ollama + nomic-embed-text + Chroma + a 22M cross-encoder, Google Colab

- 0 Block 0 · the baseline revisited, every wrong row in a box
- 1 Block 1 · HyDE
- 2 Block 2 · Hybrid BM25 + RRF
- 3 Block 3 · Reranking
- 4 Block 4 · CRAG
- 5 Block 5 · the table: which fix paid for what it cost

Exercise · the fifth variant

- Pick two families and change one setting in each
- HyDE: 3 paragraphs · Hybrid: RRF constant, pool size · Rerank: candidates, score cut · CRAG: grader prompt
- For each question where your version and the lesson's differ, say why
- Run the baseline 3 times and list the questions that change
- Two pages. Keep the CSVs.

github.com/ivanovitchm/aiengineering

Agosto - 2026						
D	S	T	Q	Q	S	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

Setembro - 2026						
D	S	T	Q	Q	S	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30			

Outubro - 2026						
D	S	T	Q	Q	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

Novembro - 2026						
D	S	T	Q	Q	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30					

Dezembro - 2026						
D	S	T	Q	Q	S	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

Unit 01

Week 01

10/08 – Re-Enroll/Planing

12/08 – Course Outline

Week 02

17/08 – Visualizing Gradient Descendent

19/08 – Rethinking the training loop

Week 03

24/08 – A simple classification problem

26/08 – From a shallow to a deep-ish classification model

Week 04

31/08 – LLM Introduction (History, Embedding)

02/09 – Architecture Overview

Week 05

07/09 – No class

09/09 – Ollama

Week 06

14/09 – RAG

16/09 – Advance RAG

Weeks 07, 08, 09, 10, 11 (Hands on Project)

21/09 – Project

23/09 – Project

28/09 – Project

30/09 – Project

05/10 – Project

07/10 – Project

12/10 – No class

14/10 – Review/Consulting

19/10 – Project Presentation (paper, slides, repo)

21/10 – Project Presentation (paper, slides, repo)

End of Unit 01 (21/10)

Unit project: advanced RAG on your own corpus

One public GitHub repo per group

```
README.md
  how to reproduce
corpus/
questions.json
notebook.ipynb
  top to bottom in Colab
results/
  baseline r1 r2 r3
  results_<technique>.csv
slides.pdf
```

F1 · Before searching

Step-Back
Multi-query / RAG-Fusion
RAG with memory

F2 · How to search

Semantic chunking
Parent-child
GraphRAG

F3 · After searching

Context compression
Ordering
Another reranker

F4 · The control loop

Adaptive RAG
Self-RAG

How the project is graded

Corpus and questions

A corpus that fits the theme; 20 well-formed questions with a verifiable answer key; the family's problem is visible in them

Baseline diagnosis

Three runs recorded; wrong rows explained with evidence from the CSV; the family's problem shown in your corpus, not just claimed

Comparison and analysis

Full table (tiers, LLM calls, latency); disagreements listed with their causes; signal told apart from noise; a conclusion that follows the data

Talk

20-minute talk that reads the examples on screen and keeps to time

Repository

The required structure; a README that lets another group reproduce your table; CSVs, paper and slides present